

■ CS Master Daily Learning Plan

Web Development (4 Weeks)

Day 1: HTML tags, headings, paragraphs, lists
Day 2: Links, images, forms
Day 3: CSS basics: selectors, colors, box model
Day 4: CSS layout: Flexbox, Grid
Day 5: JS basics: variables, types, operators
Day 6: JS functions, loops, arrays
Day 7: Mini project: Simple portfolio page
Day 8: DOM manipulation
Day 9: Event handling
Day 10: JS objects and arrays deep dive
Day 11: Fetch API / AJAX basics
Day 12: ES6+ features: arrow functions, destructuring, modules
Day 13: Project: Interactive portfolio features
Day 14: Review + small exercises
Day 15: React setup, JSX basics
Day 16: Components, props
Day 17: State, events
Day 18: Conditional rendering, lists
Day 19: Forms & controlled components
Day 20: React Router basics
Day 21: Project: Multi-page portfolio site
Day 22: Node.js setup, Express basics
Day 23: Routing, middleware
Day 24: MongoDB / MySQL basics
Day 25: CRUD operations
Day 26: Authentication (JWT / sessions)
Day 27: Connect frontend + backend
Day 28: Complete small full-stack project

AI & Data Science (4 Weeks)

Day 1: Linear Algebra basics (vectors, matrices)
Day 2: Probability & Statistics basics
Day 3: Python for Data Science: NumPy, Pandas
Day 4: Data cleaning & visualization with Matplotlib/Seaborn
Day 5: Exploratory Data Analysis (EDA)

Day 6: Machine Learning basics: regression, classification
Day 7: Implement Linear Regression from scratch
Day 8: Implement Logistic Regression from scratch
Day 9: Decision Trees & Random Forest
Day 10: K-Nearest Neighbors, SVM
Day 11: Model evaluation metrics
Day 12: Project: Predict dataset outcomes
Day 13: Deep Learning basics (Neural Networks)
Day 14: TensorFlow / PyTorch introduction
Day 15: Build a simple neural network
Day 16: CNN basics (images)
Day 17: RNN basics (sequences)
Day 18: Project: Image/Text classification
Day 19: Advanced ML topics: Hyperparameter tuning
Day 20: Project: Complete ML model
Day 21: Review + Practice exercises
Day 22: Work on Kaggle competition
Day 23: Big Data basics (Spark, Hadoop)
Day 24: Data pipelines and cleaning
Day 25: Project: Big Data mini-project
Day 26: Review and document learnings
Day 27: Prepare portfolio showcase
Day 28: Finalize AI/Data Science projects

DevOps (3 Weeks)

Day 1: Linux basics & shell scripting
Day 2: Git & GitHub version control
Day 3: Docker basics, containers setup
Day 4: Docker: images, volumes, networking
Day 5: Kubernetes basics, pods, deployments
Day 6: CI/CD concepts & tools (Jenkins, GitHub Actions)
Day 7: Project: Deploy simple app with Docker
Day 8: Project: Setup CI/CD pipeline
Day 9: Cloud basics (AWS/GCP)
Day 10: Deploy app to cloud instance
Day 11: Monitoring & Logging basics
Day 12: Infrastructure as Code basics (Terraform)
Day 13: Project: Complete DevOps pipeline
Day 14: Review & document learnings
Day 15: Advanced: Scaling & load balancing
Day 16: Security in DevOps basics

Day 17: Project: Final DevOps project

Cybersecurity (3 Weeks)

Day 1: Networking basics: TCP/IP, OSI model

Day 2: Linux for pentesting basics

Day 3: Cryptography basics: symmetric/asymmetric

Day 4: Web security basics: SQLi, XSS, CSRF

Day 5: Kali Linux tools intro: Nmap, Wireshark

Day 6: TryHackMe: basic challenges

Day 7: Penetration testing methodology

Day 8: Web app pentesting lab

Day 9: Capture the Flag (CTF) beginner challenge

Day 10: Network security concepts

Day 11: Advanced: Ethical hacking techniques

Day 12: Project: Pentesting lab report

Day 13: Secure coding practices

Day 14: Cloud security basics

Day 15: Final review & cybersecurity project